

Vial Wash Line Jam Detection System

Problem

The vial washline at Veolia, Boulder is a semi-autonomous process that inputs unpacked, dirty vials and feeds them through multiple sanitation stages before packaging. Due to the layout of the cleanroom this process inhabits, the line operators had to leave their workstation to inspect the rinse machine, which is the first and most important step in this process. The goal of this project was to provide a detect-and-alert system for the washline operators.

Process

I was tasked with designing a bracket for and finding an appropriate location to mount the pre-purchased Keyence IV4 vision sensor. After spending a week working with the wash line operators, I decided to mount the sensor in the rinse machine itself, in a corner location just outside the potential splash zone. I then designed and fabricated the bracket to screw directly into the framing of the rinse machine, and programmed the sensor's built-in AI tools to output OK/NG on a trigger-by-trigger basis.

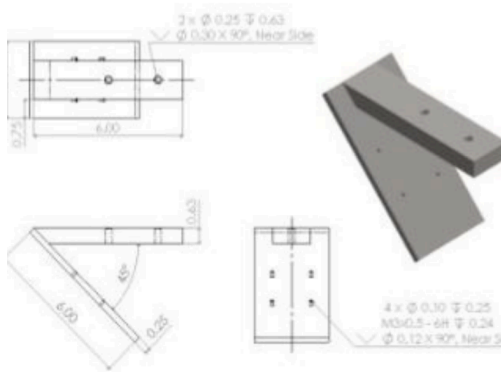


Figure 1: Bracket Dimensioned Drawing

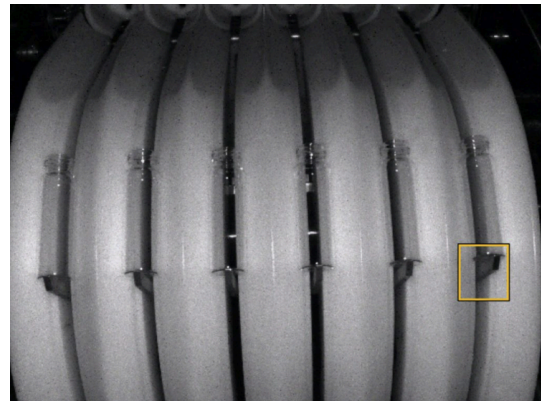


Figure 2: Sensor AI Trigger Box

Product

The camera mounting bracket was fabricated using two leftover pieces of 304 stainless-steel stock, avoiding the need to purchase additional material. The pieces were cut and machined to the required dimensions using a horizontal band saw and milling machine, then welded together to form the completed bracket. The finished bracket and camera were subsequently installed in the rinse machine as part of the wash line sensor system. The final sensor outputs the AI count tool's integer reading (0-6) to the line operator's interface to display the number of lanes with jams, and the frequency of these lane jams is currently being collected for future analysis.

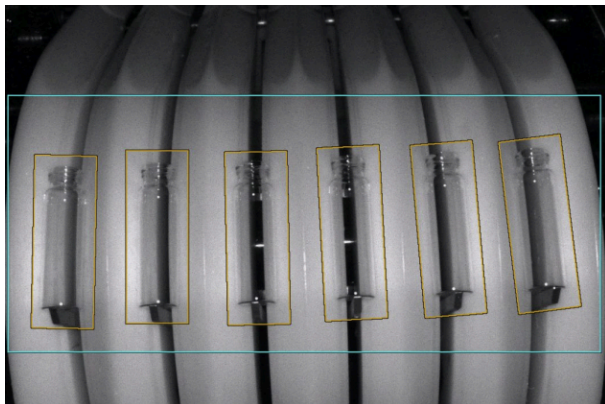


Figure 3: Sensor AI Count Tool

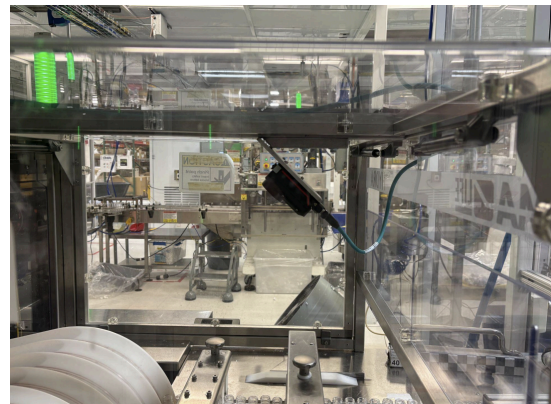


Figure 4: Installed Sensor and Bracket